

Lightweight Gated MRI–PET Fusion for Unsupervised Domain Adaptation in Alzheimer’s Disease Classification

Syed Muhammad Abdullah¹, Shahab Ud Din¹, Muhammad Haroon Shakeel²
and Murtaza Taj¹

¹ LUMS

² Adelaide University

Abstract. Multimodal neuroimaging models for Alzheimer’s disease (AD) classification often suffer from domain shift, where models trained on one cohort fail to generalize effectively to unseen clinical populations. Addressing this challenge is critical for the deployment of reliable AI systems in real-world healthcare settings. In this work, we investigate cross-domain AD versus cognitively normal (CN) classification using paired structural MRI and PET data from ADNI1 and ADNI2. We first establish a strong multimodal domain-adaptation baseline based on dual 3D CNN encoders, adversarial domain alignment, and attention-based MRI–PET fusion. Building on this foundation, we propose a novel lightweight gated fusion framework that integrates modality-specific representations through adaptive gating while incorporating weighted focal loss and adversarial adaptation to improve robustness under class imbalance and domain shift. Extensive cross-domain experiments demonstrate the effectiveness of the proposed approach. On the challenging ADNI1→ADNI2 transfer task, our method substantially improved target-domain accuracy from 38.67% to 66.67%, highlighting the benefits of adaptive multimodal fusion for domain generalization. These results demonstrate that lightweight gated MRI–PET fusion can achieve competitive or superior cross-domain performance while avoiding the computational overhead of more complex attention-based fusion mechanisms. The study contributes a practical and efficient multimodal domain-adaptation framework and provides new insights into the role of adaptive modality fusion for improving the generalizability of neuroimaging-based AD classification across heterogeneous datasets.

Keywords: Alzheimer’s disease · MRI–PET fusion · domain adaptation · gated fusion

1 Introduction

Alzheimer’s disease (AD) is a neurological disorder which causes memory loss, impact on cognitive function, and changes in everyday behavior. Reliable and early diagnosis plays a vital role in helping patients gain access to timely clinical intervention. Deep learning techniques are increasingly employed in medical

image analysis tasks, including AD classification. MRI and PET are the most common medical imaging modalities that can be combined in order to yield complementary information. While MRI represents the structural characteristics of the brain, PET shows the metabolism or functionality. Hence, MRI-PET integration allows extracting more useful information than using either MRI or PET separately [2,5].

A major challenge in medical imaging-based deep learning is out-of-distribution (OOD) generalization. In practice, a model trained on one data distribution may not perform reliably when evaluated on another distribution. This may be due to differences between training and testing data in data acquisition procedures, processing algorithms, population studied. A classifier which is able to perform well in the source domain but cannot generalize well in the target domain cannot be considered robust enough. Hence, this study focuses on cross-domain MRI-PET-based AD classification in both ADNI1→ADNI2 and ADNI2→ADNI1 transfer settings.

We approach this task by investigating binary classification AD/CN with cross-domain adaptation, where CN means cognitively normal patients. We explore two cross-domain scenarios, ADNI1 → ADNI2 and ADNI2 → ADNI1, where one domain is treated as the labeled source domain and the other as the unlabeled target domain. The unlabeled target data is utilized for domain adaptation, whereas target labels are saved exclusively for performance evaluation.

As a baseline, we used the MM-DDA multimodal domain adaptation model by Fu et al. [2]. The multimodal classifier uses individual 3D CNN encoders, multi-head attention-based fusion, and an adversarial domain adaptation mechanism.

We also perform comparisons with Dynamic Adversarial Adaptation Network (DAAN), which is a domain adaptation approach introduced by Yu et al. [7]. DAAN uses adversarial adaptation to obtain domain-invariant features, and it dynamically adjusts between global domain adaptation and class-wise domain adaptation. In our experiments, we implemented DAAN using the publicly available code and modified the data preprocessing part for our paired ADNI MRI-PET data.

We propose a lightweight gated MRI-PET fusion model for cross-domain AD classification. It involves replacing attention-based fusion with gated MRI-PET fusion where the neural network learns MRI and PET weightings for every sample. The primary focus of our research is to evaluate a baseline attention-based model that fuses multiple modalities in the context of domain adaptation with respect to a simpler gated MRI-PET fusion scheme under similar constraints of limited paired data availability. For a comparable evaluation, both methods are used for training and testing on the same MRI-PET pair data, preprocessing protocol, image-readability checks, and transfer conditions. The major contribution of this research is proposing a simple, less complex gated MRI-PET fusion method as an alternative approach to attentional fusion.

2 Methodology

2.1 Problem Formulation

This study investigates domain adaptation for Alzheimer’s disease classification using paired structural MRI and metabolic PET scans. The problem is defined as the binary classification problem of AD versus CN under domain transfer between ADNI1 and ADNI2. In particular, AD stands for the class of Alzheimer’s disease and CN stands for cognitively normal. We evaluate two transfer settings: ADNI1→ADNI2 and ADNI2→ADNI1. In each setting, the first domain is treated as the labeled source domain, while the second domain is treated as the unlabeled target domain. Let the source domain and target domain be denoted by S and T , respectively. Each source-domain sample consists of an MRI volume, a PET volume, and a diagnostic label:

$$D_S = \{(x_i^{mri}, x_i^{pet}, y_i)\}_{i=1}^{N_S},$$

where $y_i \in \{0, 1\}$ represents the binary CN/AD label. The target domain contains paired MRI and PET volumes:

$$D_T = \{(x_j^{mri}, x_j^{pet})\}_{j=1}^{N_T}.$$

In this case, labels from the target domain do not participate in training at all; their role is restricted to final assessment purposes. In other words, the model is trained on the labeled data from the source domain and unlabeled data from the target domain, with the idea of creating a feature space which can still be used for AD/CN diagnosis despite the change of the domain, from ADNI1 to ADNI2 [2].

2.2 Dataset Construction, Pairing, and Preprocessing

The MRI and PET data utilized were obtained from ADNI [1]. Because the main task is binary AD/CN classification, MCI patients were filtered out from the datasets used in the main experiments.

MRI and PET metadata were obtained separately for each ADNI domain. As ADNI includes several visits for many participants with repeated scanning, the matching process between MRI and PET data was carried out based on a subject level rather than a precise matching between acquisitions dates. For each participant who has MRI and PET scan in the metadata, all possible combinations of MRI and PET were taken into account. The first requirement for the MRI–PET match was identical AD/CN status according to both metadata. After filtering out the candidates using this criterion, the combination with the minimal absolute difference between acquisition dates was chosen. However, MRI–PET combination was selected only if its gap was no more than 180 days. That means that scans from the same date were used for matching; however, if such scans did not exist, another option with the same limitation of maximum 180-day gap was used. In this way, no more than one pair was selected for each participant.

Images which were marked as unreadable, scan folders which were missing, and those images which cannot be associated with metadata, respectively, were filtered out. Filtering these elements is very crucial, since train-validation-test splitting beforehand can give biased results since some samples will not participate at all. Thus, train-validation-test splitting was done solely after creating the usable MRI-PET paired dataset. The split into training, validation, and test datasets occurred after building the MRI-PET dataset with pairs. For the generation of pairs, only one MRI-PET pair per subject was chosen, based on the most suitable valid pair that meets the requirement of a good pair as mentioned earlier. Consequently, splitting was carried out based on paired subjects, meaning that no subject was allowed to be in two different training, validation, or test datasets.

After MRI-PET combination, each chosen scan was imported as a 3D volume and then went through the same pre-processing procedure before splitting of the data. The import process accepted various neuroimaging file formats and converted each scan to one 3D floating-point volume. Non-finite numbers like NaN and infinity were replaced with zeros. Scans that could not be imported, folders containing scans that could not be found, corrupted metadata links, timeouts and unprocessable volumes were removed from the dataset.

The intensities for MRI were strongly clipped within the 0.5th and 99.5th percentiles and normalized by z-score normalization on a per-volume basis within the foreground volume, estimated from the voxels that had more than 20th percentile intensity values. The intensities of PET images were strongly clipped within the 0.5th and 99.5th percentiles and normalized to $[0, 1]$ using min-max normalization. No SUV or SUVR was calculated. To maintain consistency within the PET images, all ADNI1 PET image entries that had descriptions including PIB, C11, or C-11 were discarded.

For scans where automatic preprocessing was not done, automatic cropping was done on foreground by considering nonzero or high-intensity voxels, where a margin of four voxels is kept around them. Light Gaussian filtering with a value of $\sigma = 1.0$ is done for such raw images. N4 bias field correction is performed optionally in preprocessing steps, but it is turned off in the experimental part. All MRI and PET scans were resampled to $113 \times 137 \times 113$ size using trilinear interpolation with `align_corners=False` setting. After resizing, modality-specific normalization was once again performed.

Initially, there were 84 candidate ADNI1 AD/CN MRI-PET pairs before the filtering stage. After removing unreadable and incomplete pairs, 58 usable ADNI1 pairs remained. In comparison, the ADNI2 dataset contained 249 usable MRI-PET pairs. A summary of the final paired datasets is presented in Table 1.

Table 1. Usable paired AD/CN MRI–PET samples after readability filtering.

Domain	Usable pairs	Train	Validation	Test
ADNI1	58	22	18	18
ADNI2	249	99	75	75

For both directions of transfer, the unlabeled target data for adversarial adaptation was the training set only. Neither the validation nor the test set from the target domain was used in the training of the feature extraction module, fusion block, classifier, and the domain discriminator. This classification was made on the basis of class labels in order to maintain the same AD/CN ratio in the training set, validation set, and test set. The final splitting percentage was around 40% for the training set, 30% for the validation set, and 30% for the test set.

2.3 Baseline Model: MM-DDA

For comparison purposes, we used the MM-DDA multimodal domain adaptation model presented in [2]. The baseline consists of three primary components: a feature encoder, a multi-head attention fusion block and an adversarial domain adaptation module.

Our baseline feature encoder has two parallel 3D CNN pipelines, each processing either MRI or PET scans. Modality-specific spatial features are extracted from the respective input volumes. Then, the feature maps generated by our MRI and PET encoders are concatenated and transformed into the sequential feature representation. This representation is fed into a multi-head attention fusion module, where cross-modality relationships are learned and a fused representation is produced. Specifically, fusion using attention is applied as it is capable of capturing interactions between multiple feature subspaces [6].

The feature representation obtained for the source domain after the fusion operation is further fed into the classifier that provides predictions about the presence of AD or CN. In addition, our baseline has a domain discriminator, which outputs whether an incoming feature representation belongs to ADNI1 or ADNI2. A gradient reversal layer is present to teach the feature encoder to generate domain-invariant features [3].

Thus, the objective function of the baseline includes the following losses:

$$\mathcal{L}_{base} = \mathcal{L}_{cls} + \mathcal{L}_{cor}^S + \mathcal{L}_{cor}^T + \mathcal{L}_{mse} + \mathcal{L}_{dom}.$$

Here, $\mathcal{L}_{cls}$ is the source-domain classification loss, $\mathcal{L}_{cor}^S$ and $\mathcal{L}_{cor}^T$ are MRI–PET correlation losses for the source and target domains, $\mathcal{L}_{mse}$ aligns the source and target fused feature representations, and $\mathcal{L}_{dom}$ is the domain discrimination loss.

2.4 Comparison Model: DAAN

Dynamic Adversarial Adaptation Network (DAAN) is one of the additional models considered for domain adaptation purpose [7]. DAAN includes a feature extractor, a label classifier, a global domain discriminator, and local class-wise domain discriminators. Global domain discriminator aids in aligning the general distribution of features from source and target domains, while the local domain discriminators help align the class-wise subdomain distributions of both domains. DAAN incorporates dynamic adversarial parameter to optimize the contributions of global and local domain adaptation.

In our experiments, we adopted the DAAN framework which is freely accessible from its official website. As DAAN was originally developed for the conventional image-based domain adaptation datasets, we customized the way of feeding the training data into the DAAN framework to ensure it could handle our paired dataset of ADNI MRI-PET images. The key components of DAAN including global and local class-wise discriminators, gradient reversal layer, as well as the dynamic adversarial weight between global and local discriminators were kept as they were. It is important to note that, in order to run their model on our dataset, we modified their publicly available code slightly but their architecture was kept same.

2.5 Lightweight Gated MRI–PET Fusion

The proposed model substitutes the expensive attention-based fusion layer with a less complex gated fusion layer. This reduces the complexity of the design while still allowing the model to learn feature-wise MRI and PET contributions for each patient.

MRI and PET volumes are first passed through separate 3D CNN encoders:

$$h_m = f_m(x^{mri}), \quad h_p = f_p(x^{pet}).$$

Global average pooling is applied to each feature map, followed by projection layers to obtain fixed-dimensional MRI and PET embeddings:

$$z_m = g_m(h_m), \quad z_p = g_p(h_p).$$

The gated fusion module receives three pieces of information:

$$[z_m, z_p, |z_m - z_p|].$$

$|z_m - z_p|$ indicates that the difference between MRI and PET embeddings in terms of features can be captured easily. This will help the gate in understanding which type of information is coming from both modalities, whether there is any discrepancy among these sources, and which source should be taken into consideration more for particular features. The gate is computed as:

$$g = \sigma(\text{MLP}([z_m, z_p, |z_m - z_p|])),$$

where $g \in [0, 1]^d$ is a feature-wise gate vector with the same dimensionality as z_m and z_p . In our implementation, $d = 128$. Therefore, the gate is not a single scalar for the entire patient; rather, each feature dimension receives its own MRI–PET weighting. The fused representation is computed feature-wise as:

$$z_f = g \odot z_m + (1 - g) \odot z_p,$$

where $\odot$ denotes element-wise multiplication. For a given feature dimension, a gate value close to 1 makes the fused representation rely more on the MRI embedding, while a value close to 0 makes it rely more on the PET embedding.

The fused representation z_f is then passed to a classifier:

$$\hat{y} = C(z_f),$$

which predicts whether the subject belongs to the CN or AD class. The architecture diagram is shown in Fig. 1.

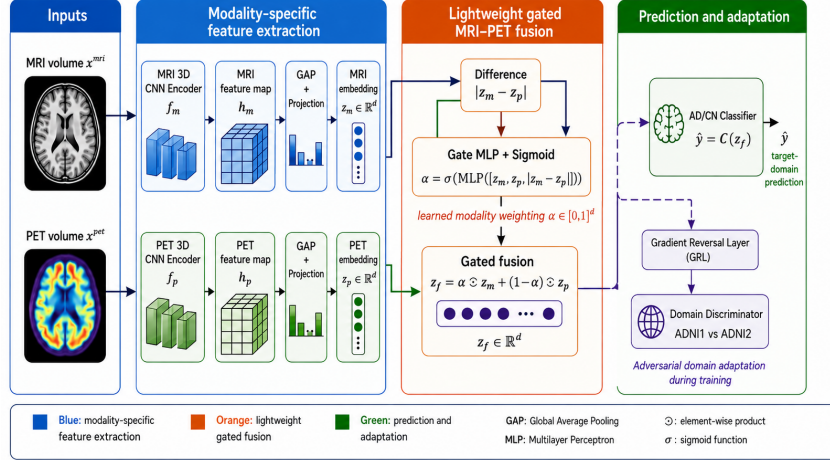


Fig. 1. Architecture of the proposed lightweight gated MRI–PET fusion model. MRI and PET volumes are processed through separate 3D CNN encoders, projected into fixed-dimensional embeddings, and combined using a learned gated fusion mechanism. The fused representation is then used for AD/CN classification and adversarial domain adaptation.

2.6 Training Objective

The proposed model is trained in two stages. The first stage performs source-only supervised training, while the second stage introduces unlabeled target-domain data for adversarial domain adaptation.

Stage 1: Source-only Supervised Training During the first step, only the labeled data from the source domain is utilized. In this step, the learning of MRI/PET feature extraction, gated fusion, and AD/CN classification takes place. Weighted focal loss is utilized for achieving stability during training in case of class imbalance. The focal loss technique was proposed for concentrating more on hard examples and not the easy ones [4]. The weighted focal loss is:

$$\mathcal{L}_{focal} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} (1 - p_{t,i})^\gamma \log(p_{t,i}),$$

where $p_{t,i}$ is the predicted probability assigned to the correct class, w_{y_i} is the class weight, and γ is the focal focusing parameter.

We also include a modality alignment loss between MRI and PET embeddings:

$$\mathcal{L}_{align} = \left\| \frac{z_m}{\|z_m\|} - \frac{z_p}{\|z_p\|} \right\|_2^2.$$

This encourages MRI and PET embeddings from the same patient to remain semantically compatible.

The Stage 1 objective is:

$$\mathcal{L}_{stage1} = \mathcal{L}_{focal} + \lambda_{align} \mathcal{L}_{align}.$$

Stage 2: Unsupervised Domain Adaptation: In the second stage, labeled source-domain samples and unlabeled target-domain samples are used together. The source-domain samples continue to train the AD/CN classifier through the supervised focal loss, while the target-domain samples are used only for domain adaptation. The fused representations from both domains are passed through a gradient reversal layer and then into a domain discriminator. The domain label is defined as

$$d = 1 \quad \text{for source}, \quad d = 0 \quad \text{for target}.$$

The domain classification loss is

$$\mathcal{L}_{dom} = -\frac{1}{N} \sum_{i=1}^N \left[d_i \log(\hat{d}_i) + (1 - d_i) \log(1 - \hat{d}_i) \right],$$

where $\hat{d}_i$ is the predicted probability that the i -th fused feature belongs to the source domain. During forward propagation, the gradient reversal layer acts as an identity function. During backpropagation, it reverses the gradient passed to the encoder and fusion module by multiplying it with $-\lambda_{GRL}$. Thus, the domain discriminator minimizes L_{dom} to distinguish ADNI1 from ADNI2, while the encoder and fusion module receive the reversed gradient and learn domain-invariant fused features [3]. In our implementation, $\lambda_{GRL} = 0.1$.

In the target domain, alignment loss is determined in an unsupervised fashion solely based on MRI and PET representations of the target individual. In

particular, (L_{align}^T) is defined as the normalized distance between (z_m^T) and (z_p^T) , exactly like in the source domain, except there is no use of the target diagnosis (y^T) at all. It implies that target-domain samples contribute to MRI–PET representation alignment and domain-invariant feature learning without using target diagnostic labels.

The overall Stage 2 objective is

$$L_{stage2} = L_{focal}^S + \lambda_{align} (L_{align}^S + L_{align}^T) + \lambda_{dom} L_{dom}.$$

Here, L_{focal}^S denotes the supervised focal loss computed only on labeled source-domain samples. L_{align}^S and L_{align}^T denote the MRI–PET alignment losses for source and target samples, respectively, and λ_{dom} controls the strength of adversarial domain adaptation. Target-domain diagnostic labels are never used during Stage 2 training.

2.7 Experimental Setup and Evaluation Metric

The MM-DDA baseline and the proposed lightweight gated MRI–PET fusion model are evaluated under two cross-domain transfer settings: ADNI1→ADNI2 and ADNI2→ADNI1. In each setting, the source domain is used with diagnostic labels during training, while the target-domain labels are not used for training and are reserved for evaluation. For domain adaptation, only the labeled training data from the target domain was used. Data from the target domain for validation and testing was not included in the model’s training process at all. Target test data was entirely unseen while training the models, and only used to evaluate their performance. Tuning of the models’ thresholds was done on the source validation split. To ensure a fair comparison, both models are trained and evaluated using the same dataset construction procedure, MRI–PET pairing criteria, preprocessing pipeline, train-validation-test split, and evaluation protocol. Any additional comparison models are also evaluated on the same limited paired dataset to avoid mixing results obtained from different datasets or different split settings. Accuracy is the primary evaluation criterion used to evaluate the three methods. Accuracy is the rate at which the algorithm predicts the correct target sample:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

The advantage of this measure is that the base-line case and the proposed solution can be directly compared within identical cross-domain scenarios.

The main hyperparameters used in the experiments are shown in Table 2.

3 Results

This section presents the evaluation structure for the cross-domain AD/CN classification experiments. The final results are reported for both transfer directions, ADNI1→ADNI2 and ADNI2→ADNI1. All models were evaluated on the same paired MRI–PET dataset described in Table 1 to ensure fairness.

Table 2. Main hyperparameters used in the experiments.

Hyperparameter	Value
Input shape	$113 \times 137 \times 113$
Batch size	1
Optimizer	Adam
Learning rate	9×10^{-7}
Dropout	0.5
Stage 1 epochs	150
Stage 2 epochs	50
GRL coefficient λ_{GRL}	0.1
Domain-loss weight λ_{dom}	0.10
Focal loss γ	2
Maximum allowed MRI–PET date gap	180 days

3.1 Cross-domain Classification Performance

Table 3 presents both the results for the ADNI1→ADNI2 and the reverse ADNI2→ADNI1 transfer setting. In the first setting, ADNI1 is used as the labeled source domain and ADNI2 is used as the unlabeled target domain. In the reverse setting, ADNI2 is used as the labeled source domain and ADNI1 is used as the unlabeled target domain. Target labels are not used during training and are reserved only for evaluation.

Table 3. Cross-domain AD/CN classification performance on the selected paired MRI–PET dataset. Higher values indicate better performance.

Transfer Direction	Method	ACC (%)
ADNI1→ADNI2	DAAN [7]	64.10
	MM-DDA [2]	38.67
	Proposed gated fusion model	66.67
ADNI2→ADNI1	DAAN [7]	59.09
	MM-DDA [2]	50.00
	Proposed gated fusion model	50.00

3.2 Accuracy Change Relative to Baseline Models

Table 4 shows the change in target-domain accuracy from the baseline models to our lightweight gated fusion model.

The strongest improvement was observed in the ADNI1→ADNI2 transfer setting, where the proposed model improved target accuracy by 28.00 percentage points as compared to MM-DDA. This is the main quantitative gain observed in the final experiments. In the ADNI2→ADNI1 direction, the proposed model matched the baseline accuracy.

Table 4. Accuracy gain of the proposed model compared with MM-DDA and DAAN baselines.

Baseline Model	Transfer Direction	Accuracy Gain
MM-DDA	ADNI1→ADNI2	+28.00%
MM-DDA	ADNI2→ADNI1	+0.00%
DAAN	ADNI1→ADNI2	+2.57%

3.3 Ablation Study

An ablation study was conducted using four model variants: CNN fusion baseline, gated fusion, gated fusion with focal loss, and the full proposed model. The aim was to assess whether performance gains came from gated fusion alone or from the combined components. The results are presented in Tables 5 and Fig. 2. For ADNI1→ADNI2, the full proposed model achieved the highest accuracy of 66.67%, outperforming the CNN baseline at 38.67% and the individual component variants at 33.33%. In contrast, for ADNI2→ADNI1, the results were less stable, with most variants achieving 50.00% accuracy and gated fusion with focal loss dropping to 40.00%.

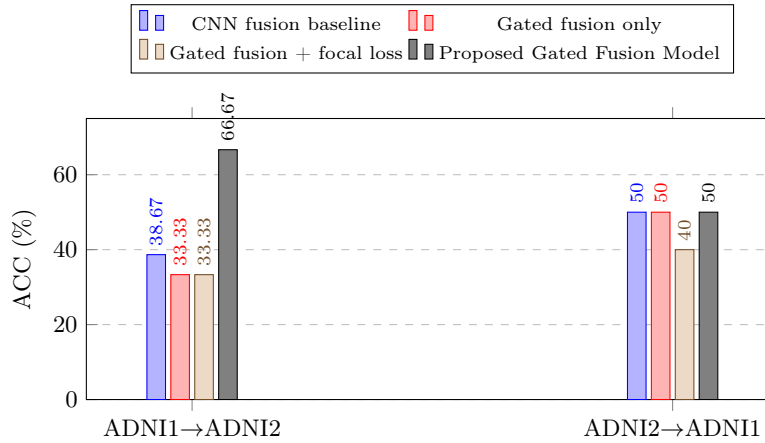
**Fig. 2.** Ablation study on target-domain accuracy (ACC) for the proposed gated MRI–PET fusion model. The chart compares the CNN fusion baseline, gated fusion only, gated fusion with focal loss, and the full proposed model across the two transfer settings.

Table 5. Ablation results for both cross-domain transfer directions on the selected paired MRI-PET dataset. Higher values indicate better performance.

Model variant	ACC (%)
ADNI1→ADNI2	
CNN fusion baseline	38.67
Gated fusion only	33.33
Gated fusion + focal loss	33.33
Proposed Gated Fusion Model	66.67
ADNI2→ADNI1	
CNN fusion baseline	50.00
Gated fusion only	50.00
Gated fusion + focal loss	40.00
Proposed Gated Fusion Model	50.00

3.4 Key Findings

The primary result is that the presented lightweight gated MRI-PET fusion model outperforms the MM-DDA baseline in terms of improving target domain accuracy in the ADNI1→ADNI2 transfer scenario while using a simpler fusion process. The baseline performance was 38.67(%). In contrast, the proposed model resulted in 66.67(%) target domain accuracy meaning the gain was 28.00 (%). Additionally, the proposed approach which has fewer parameters, as can be seen in Table 6 than DAAN model outperformed it in terms of increasing target domain accuracy in this direction by 2.57 percentage points: DAAN achieved 64.10 (%).

In terms of the reverse transfer from ADNI2 to ADNI1, the proposed model and MM-DDA yielded 50.00 (%) accuracy. Hence, the proposed model had similar performance compared to the baseline with a simpler architecture. DAAN performed better in this direction but, this outcome needs to be viewed with the fact that our proposed model has much fewer parameters than DAAN as can be seen in Table 6.

In general, the findings validate the primary motivation behind this research. On the one hand, the suggested gated fusion approach is much simpler than the corresponding MM-DDA attention-based architecture and shows better performance in terms of the transfer from ADNI1 to ADNI2, while showing similar results in terms of transferring in the opposite direction. At the same time, although the suggested fusion framework did not outperform DAAN in one case, but it outperformed in the other case while having a simpler, less complex architecture.

Table 6. Number of trainable parameters for all the models. Lower values indicate better model.

Model	Trainable Parameters
DAAN	1,414,776
MM-DDA	1,229,428
Proposed Gated Fusion Model	1,032,052

4 Discussion

4.1 Comparison with the MM-DDA Baseline

The MM-DDA model is a robust approach for multimodal domain adaptation due to its use of 3D CNN for feature extraction, attention-based integration of MRI-PET modalities, and adversarial domain adaptation [2]. Nonetheless, The MM-DDA baseline uses a more complex fusion design because it relies on multi-head attention in addition to adversarial domain adaptation.

The proposed model addresses this issue by replacing attention-based fusion with a gated MRI-PET fusion mechanism., where, instead of focusing on the attention learning task, the model determines the importance of MRI and PET in each case of fusion. It is simpler and less complex than attention as can be seen in Table 6 and yet preserves information from multiple modes. The ADNI1→ADNI2 results suggest that improved target-domain accuracy can be achieved using a simpler fusion model.

In the case of ADNI2→ADNI1, although the proposed model showed same accuracy in comparison to MM-DDA as a baseline model, but the proposed model achieved that accuracy with fewer parameters.

4.2 Comparison with DAAN

DAAN was considered as one more method proposed by Yu et al. [7] for comparison within domain adaptation. In the transfer from ADNI1 to ADNI2, the accuracy reached by DAAN was 64.10 (%), whereas the accuracy obtained by the proposed method was 66.67 (%) even with fewer parameters. This indicates the fact that the proposed gating mechanism in MRI-PET fusion model outperformed DAAN in this case. This result suggests that the proposed model can remain competitive with existing more complex domain-adaptation methods in the ADNI1→ADNI2 setting.

In the ADNI2→ADNI1 case, DAAN had an accuracy rate of 59.09 (%), beating both the MM-DDA benchmark and the proposed model, whose accuracy was 50.00 (%). However, the model complexity of DAAN is much greater than the proposed model as can be seen in Table 6.

4.3 Interpretation of the Ablation Study

Insights on how the suggested components work can be drawn from the ablation experiments. While using ADNI1→ADNI2 for the target domain, the CNN

Fusion Baseline model had an accuracy of 38.67 (%). The gated fusion model obtained an accuracy of 33.33 (%), while the combination of gated fusion and focal loss gave an accuracy of 33.33 (%). The results indicate that replacing the fusion model or applying focal loss alone could not help boost the performance of the target domain.

This is an important observation since this indicates that the combination of gated fusion, focal loss, and domain adaptation is best applied together. Gated fusion is a light method for combining the information from both modalities, while focal loss can help solve issues related to class imbalances during training [4], and domain adaptation pushes the fused representation towards being domain-invariant [3].

As far as the ADNI2→ADNI1 ablation is concerned, it was not conclusive enough. All three models – fusion CNNs baseline, gated fusion alone, and proposed complete architecture – have 50.00 (%) as their accuracy, whereas focal loss with gated fusion had 40.00 (%). Hence, it can be seen again that the reverse transfer direction was less stable, likely due to the small ADNI1 target test set.. Thus, The ablation results suggest that the full combination of gated fusion, focal loss, and adversarial adaptation is more effective than using the individual components alone.

4.4 Implications for Lightweight Multimodal Domain Adaptation

These findings imply that lightweight multimodal fusion is a better option as compared to attention-based fusion in scenarios where the amount of paired data is limited. The acquisition of paired scans in applications from the field of medical imaging tends to be challenging, and much of the collected data will be excluded after pairing. Under such circumstances, simplifying the model is preferable. A less complex network may train better, generalize better, and remain stable when there are relatively few available MRI–PET pairs.

Hence, the proposed gated fusion approach is not only helpful as a high-performing network, but as an example design in general. They indicate that multimodal domain adaptation does not necessarily lead to increasing complexity of the fusion component, but a well-defined and interpretable gating operation can achieve comparable performance without becoming more complex.

4.5 Limitations and Future Work

The main limitation of this study is the small number of usable paired MRI–PET cases, especially in ADNI1. The number of final readable cases was limited. The second setting (ADNI2 → ADNI1) was most affected by the low amount of targets in the test set since ADNI1 test data was not large.

The other limitation is that experiments were done using ADNI1 and ADNI2 only. While these data sets are adequate for research within domains of ADNI, it will be helpful to use other data sets, such as AIBL or OASIS to check whether the suggested approach can generalize for different kinds of data.

5 Conclusion

The paper examined cross-domain adaptation in AD classification by using paired MRI and PET images obtained from ADNI1 and ADNI2 data sets. The problem was considered in terms of binary AD/CN classification where two transfer tasks were performed: ADNI1 $\rightarrow$ ADNI2 and ADNI2 $\rightarrow$ ADNI1. For ensuring the validity of experiments, all the evaluated models had been trained and tested on the same preselected paired MRI–PET images.

The key contribution of the study is the proposed lightweight gated MRI–PET fusion technique which serves as an alternative solution to the attention-based baseline approach similar to MM-DDA models. Unlike the latter, the proposed method does not use multihead attention and instead learns gated MRI and PET contributions per each sample.

In the ADNI1 $\rightarrow$ ADNI2 domain adaptation problem, the proposed model resulted in an improvement in terms of target domain accuracy. The proposed model achieved an accuracy of 66.67(%) while the MM-DDA achieved 38.67(%) and DAAN achieved an accuracy of 64.10(%). In the reverse ADNI2 $\rightarrow$ ADNI1 domain adaptation problem; both models resulted in the same level of accuracy which was equal to 50.00(%).

However, one of the key limitations of this research was the relatively low number of paired MRI–PET examples available for ADNI1 data set. In order to overcome the stated limitation, future research could test the approach on bigger multimodal data sets and use additional external data sets.

References

1. Alzheimer’s disease neuroimaging initiative (adni). <https://ida.loni.usc.edu> (2026), accessed for MRI and PET neuroimaging data
2. Fu, B., Shen, C., Liao, S., Wu, F., Liao, B.: Prediction of alzheimer’s disease based on multi-modal domain adaptation. *Brain Sciences* **15**(6), 618 (2025). <https://doi.org/10.3390/brainsci15060618>
3. Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., Lempitsky, V.: Domain-adversarial training of neural networks. *Journal of Machine Learning Research* **17**(59), 1–35 (2016)
4. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: *Proceedings of the IEEE International Conference on Computer Vision*. pp. 2980–2988 (2017)
5. Liu, Y., Fan, L., Zhang, C., Zhou, T., Xiao, Z., Geng, L., Shen, D.: Incomplete multi-modal representation learning for alzheimer’s disease diagnosis. *Medical Image Analysis* **69**, 101953 (2021). <https://doi.org/10.1016/j.media.2020.101953>
6. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: *Advances in Neural Information Processing Systems*. vol. 30 (2017)
7. Yu, C., Wang, J., Chen, Y., Huang, M.: Transfer learning with dynamic adversarial adaptation network. In: *2019 IEEE International Conference on Data Mining (ICDM)*. pp. 778–786. IEEE (2019). <https://doi.org/10.1109/ICDM.2019.00088>